Open X-Embodiment · arXiv 2310.08864 · Nov 2 · Data

Empiricist

11+3

Mix-cardinality smoke · GPU honesty
Bridge-only vs +k · leave-Bridge · underfit probe

Paper club · Data cluster · due 9:30 AM PT

Spine: pool ≠ train mix · capacity×mix · steal mix science, don’t drink the river

Open X-Embodiment Collaboration · 21 institutions · robotics-transformer-x.github.io

Empiricist talk arc · 11+3

0:00	2:00	5:00	7:30	9:30	11:00
Tattoos	**P0 matrix**	**Leave-Bridge**	**Underfit probe**	**Defer list**	**Q&A**
headline %	smoke + mix-card	Partial moral	capacity×mix	no RT-X retrain	3 min

Spine: pool ≠ train mix · capacity×mix · steal mix science, don’t drink the river

Empiricist tattoos — memorize

Pool

22 emb · 60 ds · 1M+
527 skills · RLDS

Train

9 manipulators
≠ full pool

Fig.4

~+50% relative
small-domain lift

Table I

RT-1-X 73% vs
RT-1 92% (underfit)

Emergent

27.3% → 75.8%
(~3×)

no-Bridge

42.8%
load-bearing

Eval

3600 trials
6 robots

Downstream

Octo ~800k
OpenVLA ~970k

Empiricist priority matrix (GPU honesty)

Pri	Experiment	Smoke	Est. GPU-h
P0 Y	RLDS shard inventory / loader smoke (1–3 ds)	Y	<1
P0 Y	Mix-cardinality: Bridge-only vs +2 vs +5 FT	Y	8–24
P0 P	Leave-Bridge-out (Table II moral)	Partial	12–40
P1	Small-domain lift proxy (Fig.4 moral)	Partial	8–20
P1	Capacity×mix underfitting check (Table I)	Y	10–30
P1	Embodiment leave-one-out	Partial	16–48
P1	Same mix → Octo / OpenVLA / π_0 FT	Partial	20–60
Defer	Full RT-1-X / RT-2-X retrain · full OXE mirror	N	≫1k

Default matrix: {bridge | bridge+k | leave_bridge} × {scratch|octo|openvla} × ~100 demos FT

Falsifiable hypotheses (Empiricist)

H-mix-lift

Curated multi-mix init/FT > single-domain on small public sets (Fig.4 moral)

H-bridge-load

Leave-Bridge-out selectively destroys Bridge-unique skill transfer (Table II)

H-underfit

Fixed small capacity + heavy heterogeneous mix lowers in-domain SR (Table I)

H-schema

Per-dataset action norms / control-mode tags beat naive shared heads

H-cardinality

Which k datasets matter more than raw k (curated > random)

H-open-successor

Value is mix cards + FT on Octo/OpenVLA/π_0 — not rebuilding RT-1-X

CLEAR: Pool ≠ RT-X train mix

Marketing blur is the #1 citation failure mode

OXE POOL

Datasets	**60**
Embodiments	**22**
Institutions / labs	**21 / 34**
Trajectories	**1M+**
Skills / tasks	**527 / 160,266**
Scenes	**311**
Schema	**RLDS TFRecords**

RT-X TRAIN MIX

Manipulators	**9 (≠ full 22)**
Sources (ex.)	**RT-1, QT-Opt, Bridge, Play, Jaco, Cable, …**
RT-1-X	**robotics mix only**
RT-2-X	**~1:1 web : robotics**
Eval	**3600 trials / 6 robots**
Downstream	**Octo ~800k · OpenVLA ~970k**

Moral: every claim needs a frozen mix card — pool size ≠ train size ≠ Octo/OpenVLA subset

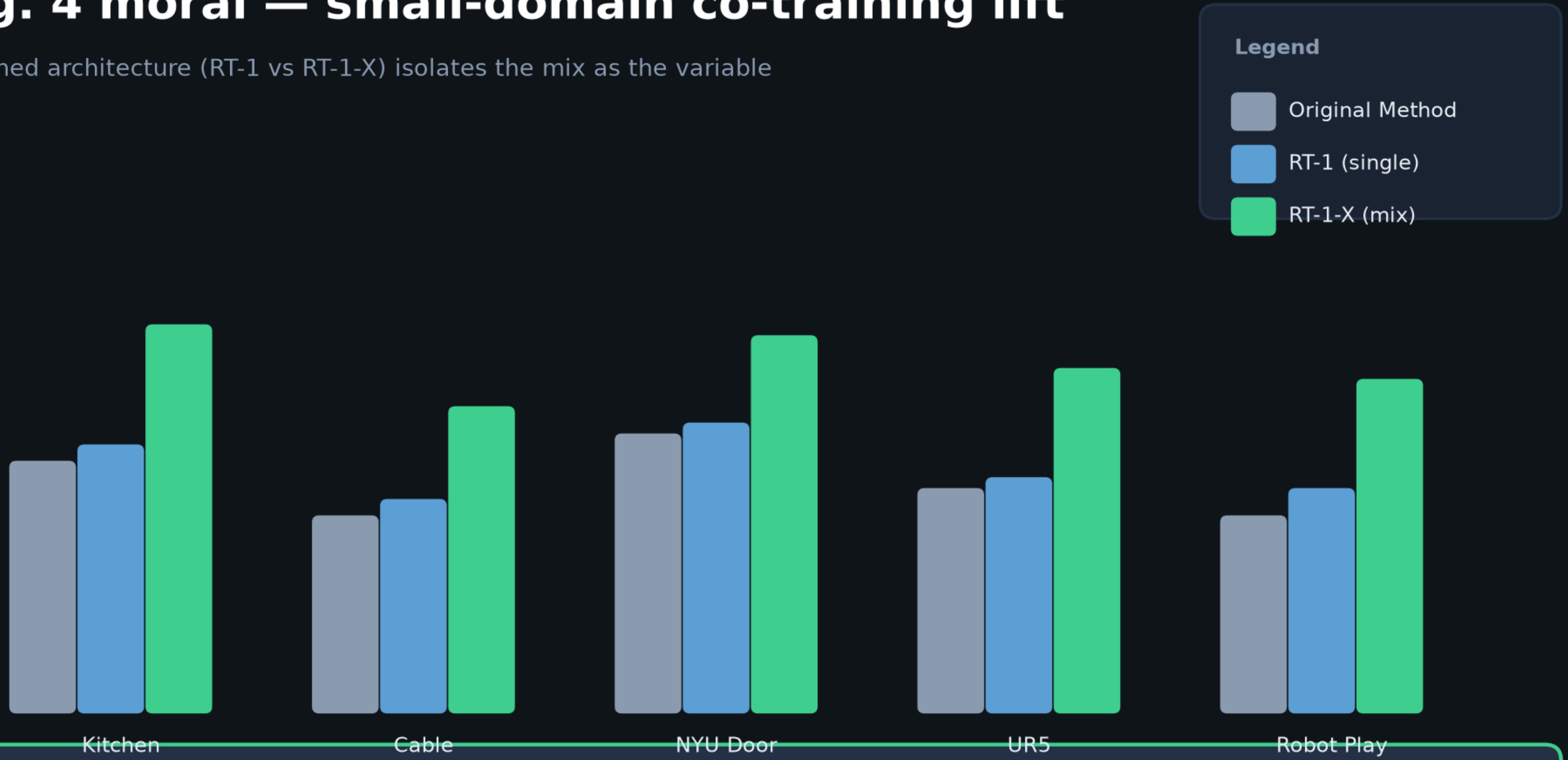
Cite “22 robots / 1M+” for the commons; cite “9 manipulators” for RT-X experiment results

Schema / consolidation used for RT-X

Observation	Image history + language; one canonical camera per dataset; unaligned poses
Action	7-DoF EE (x,y,z,r,p,y,grip) — abs/rel/vel as in source; NOT frame-aligned
Discretization	256 bins/dim · 8 outputs = 7 EE + episode terminate
Normalization	Per-dataset before binning; de-norm is embodiment-specific
Gap module	NONE — shared policy eats the soup; transfer is learned
RT-1-X	35M FiLM-EfficientNet + Transformer · robotics mix only · 3–10 Hz local
RT-2-X	PaLI-X VLA · actions as text tokens · ~1:1 web:robot · 5B/55B

Fig. 4 moral — small-domain co-training lift

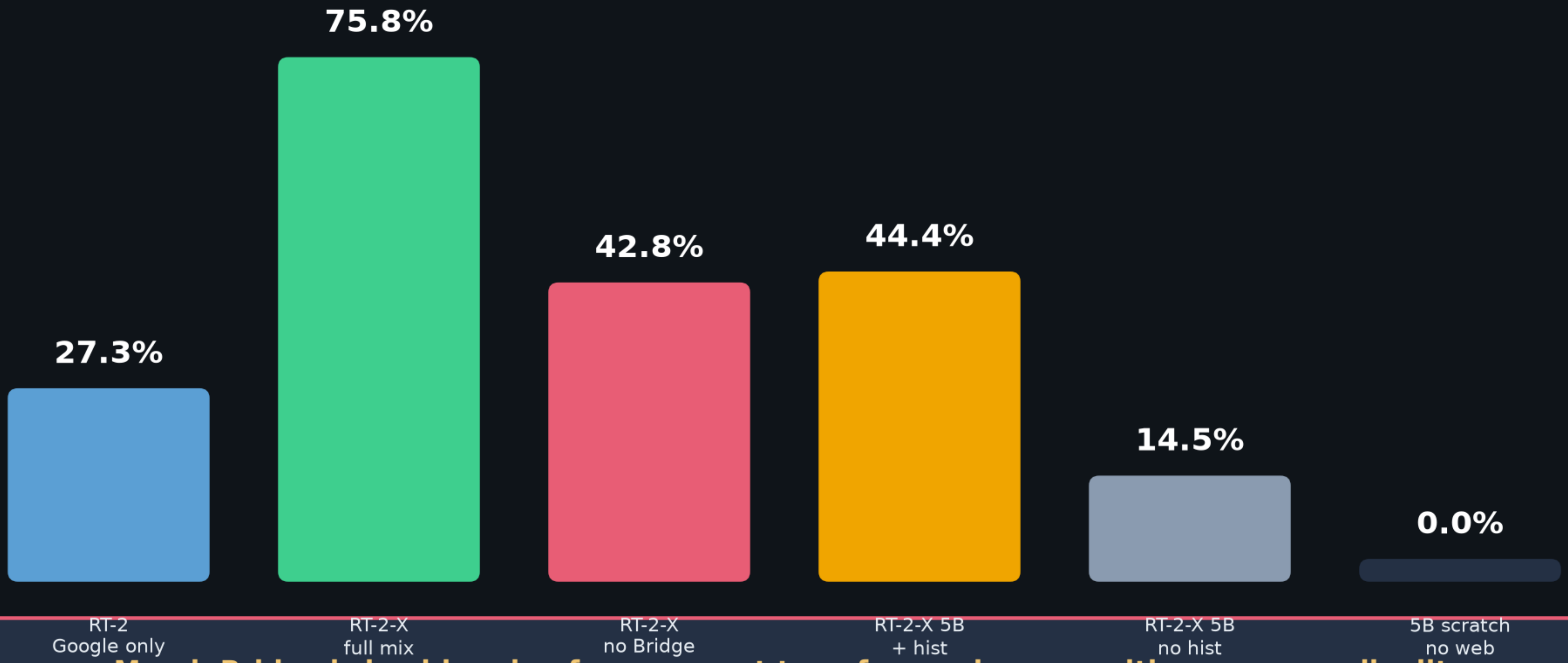
Matched architecture (RT-1 vs RT-1-X) isolates the mix as the variable



CLEAR headline: RT-1-X mean success ~50% higher than Original or single-domain RT-1

Table II — Bridge ablation (emergent skills)

Bridge→Google Robot skills absent from Google-only data



Moral: Bridge is load-bearing for emergent transfer — mix composition > raw cardinality

Capacity × mix co-design (Table I moral)

More data is not free — small models underfit rich X-mixes on large native domains

RT-1-X (35M) on X-mix

Google 6-skill: 73% (vs RT-1 92%)
Bridge IRIS: 27% (vs RT-1 40%)

Underfits large native sets

RT-2-X (55B) on X-mix

Google 6-skill: 91% (recovers)
Bridge IRIS: 50% (beats)

Capacity absorbs diversity

BEHAVIOR implication

- If shipping $\pi_{0.5}$ / GR00T-scale → you can afford OXE-like width (with weight table).
- If shipping a tiny policy → curate narrower; gate external shards behind validation q-score.
- Empiricist underfit probe: same heavy mix, small vs base open ckpt, matched steps.

Table II — Emergent skills / Bridge ablation / capacity

(1)	RT-2	55B	none	Google Robot action	Yes	Web-pretrained	27.3%	**62%**
(2)	RT-2-X	55B	none	Robotics data	Yes	Web-pretrained	**75.8%**	**61%**
(3)	RT-2-X	55B	none	Robotics data except Bridge	Yes	Web-pretrained	42.8%	54%
(4)	RT-2-X	5B	2	Robotics data	Yes	Web-pretrained	44.4%	52%
(5)	RT-2-X	5B	none	Robotics data	Yes	Web-pretrained	14.5%	30%
(6)	RT-2-X	5B	2	Robotics data	No	From scratch	0%	1%
(7)	RT-2-X	5B	2	Robotics data	No	Web-pretrained	48.7%	47%

TABLE II: Ablations to show the impact of design decisions on generalization (to unseen objects, backgrounds, and environments) and emergent skills (skills from other datasets on the Google Robot), showing the importance of Web-pretraining, model size, and history.

B. *Improved generalization to out-of-distribution settings*

We now examine how X-embodiment training can enable better generalization to out-of-distribution settings and more complex and novel instructions. These experiments focus on the high-data domains, and use the RT-2-X model.

Unseen objects, backgrounds and environments. We first conduct the same evaluation of generalization properties as proposed in [9], testing for the ability to manipulate unseen objects in unseen environments and against unseen backgrounds. We find that RT-2 and RT-2-X perform roughly on par (Table II, rows (1) and (2), last column). This is not unexpected, since RT-2 already generalizes well (see [9]) along these dimensions due to its VLM backbone.

Emergent skills evaluation. To investigate the transfer of knowledge across robots, we conduct experiments with the Google Robot, assessing the performance on tasks like the ones shown in Fig. 5. These tasks involve objects and

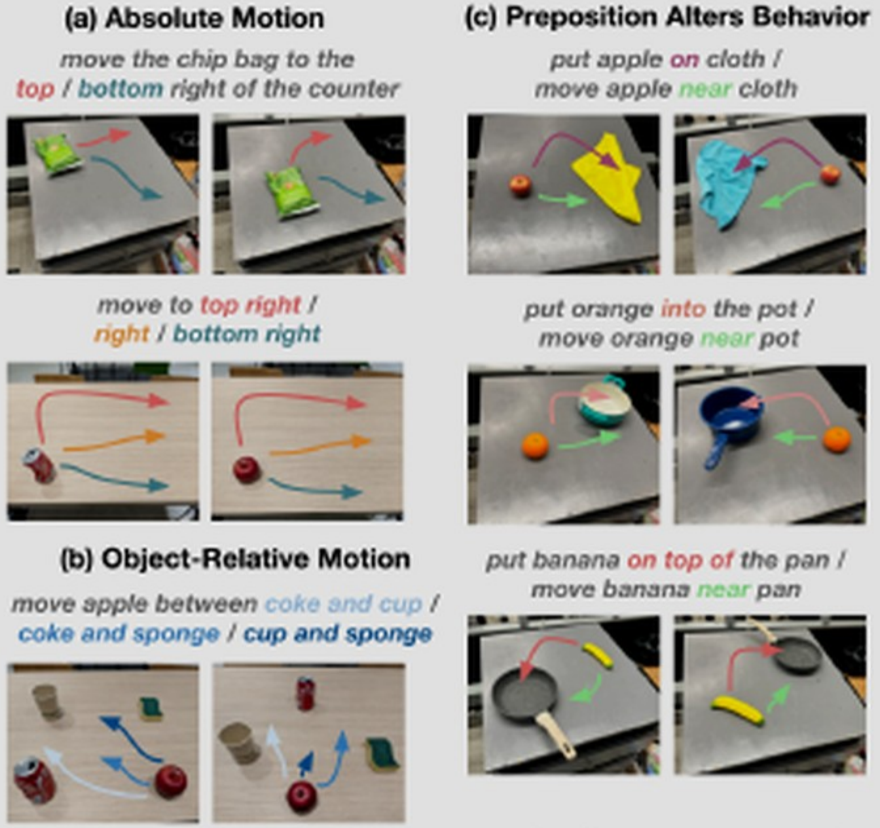


Fig. 5: To assess transfer between embodiments, we evaluate the

Table I — Large-data: RT-1-X underfits (73% vs 92%)

strong performance in these two evaluation scenarios.

Table [91] datasets. RT-1-X is trained on only robotics mixture data defined above, whereas RT-2-X is trained via co-fine-tuning (similarly to the original RT-2 [9]), with an approximately one to one split of the original VLM data and the robotics data mixture. Note that the robotics data mixture used in our experiments includes 9 embodiments which is fewer than the entire Open X-Embodiment dataset (22) – the practical reason for this difference is that we have continued to extend the dataset over time, and at the time of the experiments, the dataset above represented all of the data. In the future, we plan to continue training policies on the extended versions of the dataset as well as continue to grow the dataset together with the robot learning community.

At inference time, each model is run at the rate required for the robot (3-10 Hz), with RT-1 run locally and RT-2 hosted on a cloud service and queried over the network.

V. Experimental Results

Our experiments answer three questions about the effect of X-embodiment training: (1) Can policies trained on our

tion and robot embodiment as in the respective publications. For large-scale dataset experiments, we consider Bridge [95] and RT-1 [8] for in-distribution evaluation and use their respective robots: WidowX and Google Robot.

For each small dataset domain, we compare the performance of the RT-1-X model, and for each large dataset we consider both the RT-1-X and RT-2-X models. For all experiments, the models are co-trained on the full X-embodiment dataset. Throughout this evaluation we compare with two baseline models: (1) The model developed by the creators of the dataset trained only on that respective dataset. This constitutes a reasonable baseline insofar as it can be expected that the model has been optimized to work well with the associated data; we refer to this baseline model as the *Original Method* model. (2) An RT-1 model trained on the dataset in isolation; this baseline allows us to assess whether the RT-X model architectures have enough capacity to represent policies for multiple different robot platforms simultaneously, and whether co-training on multi-embodiment data leads to higher performance.

P0: RLDS smoke + mix-cardinality + leave-Bridge moral
Tattoo: +50% · 73 vs 92 · 27.3→75.8 · no-Bridge 42.8

Subset FT on Octo/OpenVLA/π_0 — never full RT-X retrain
Log mix card ID + GPU/disk next to every SR

Q&A · keep the spine clear

Open X-Embodiment

Robotic Learning Datasets and RT-X Models

arXiv 2310.08864 · Nov 2 Data cluster

OXE = municipal water · BEHAVIOR’26 = bottled specialty blend

Steal mix science — don’t drink the river

Pool: 60 ds / 22 emb / 1M+ traj   Train mix: 9 manipulators   Eval: 3600 trials / 6 robots

Headlines: Fig.4 ~+50% · Table I RT-1-X 73% vs RT-1 92% · emergent 27.3%→75.8% · no-Bridge 42.8%

Seat 1 Visionary = Kanta Saito · Quiet build